Demonstration of Proposed Features

Date	15 July 2026
Team ID	SWTID-2026-3798
Project Name	Rising Waters-Flood Prediction System
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrate d (Yes / No)	Remarks
1	Data Collection and Dataset Integration	Collected historical weather dataset (rainfall,cloud,visibility,seasonal patterns) for flood prediction	Implemented	Yes	Dataset sourced and verified for completeness
2	Exploratory Data Analysis and Visualization	Performed univariate,multivariate,and description analysis to understand feature relationships	Implemented	Yes	Visualizations included in Jupyter notebook
3	Data preprocessing	Handled missing values,outliers,categorical encoding,train-test split,and feature scaling	Implemented	Yes	Clean dataset ready for model training
4	Multi-Model Building(Decision Tree,Random forest,KNN,XGBoost)	Trained four classification models and compared performance metrics	Implemented	Yes	XGBoost selected as best model with 96.55% accuracy
5	Model Evaluation and Saving	Evaluated models on test data and saved the best-performing model for deployment	Implemented	Yes	Model serialized for use in Flask app

6	Flask Web Application	Built a web interface where users input rainfall/visibility data to get real-time flood risk predictions	Implemented	Yes	Deployed locally;IBM Cloud deployment planned/pending

Feature Implementation Summary:

Total Features Proposed	**6**
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%